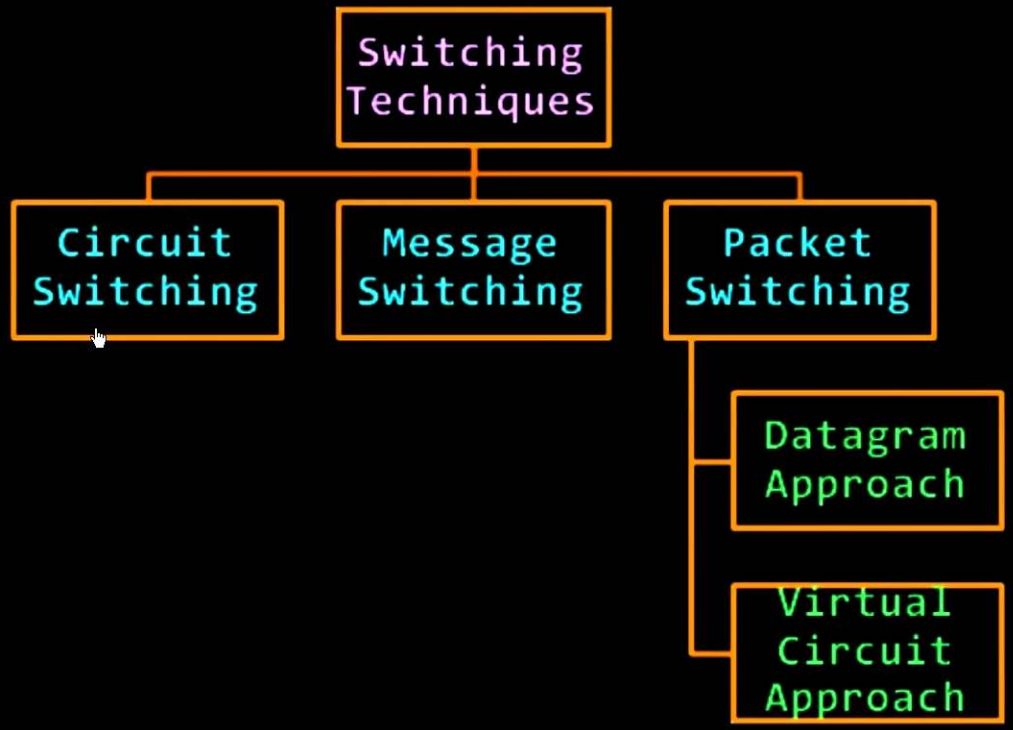
Switching techniques



Circuit switching

★ A dedicated path is established between the sender and receiver.

★ Before data transfer, connection will be established first.

★ Example: Telephone network.

3 phases in circuit switching:

1. Connection establishment.
2. Data transfer
3. Connection Disconnection.



Message switching

- ★ Store and forward mechanism.
- ★ Message is transferred as a complete unit and forwarded using store and forward mechanism at the intermediary node.
- ★ Not suited for streaming media and real-time applications.



Packet switching

- ★ The internet is a packet switched network.
- ★ Message is broken into individual chunks called as packets.
- ★ Each packet is sent individually.
- ★ Each packet will have source and destination IP address with sequence number.
- ★ Sequence numbers will help the receiver to
  - ○ Reorder the packets.
  - ○ Detect missing packets and
  - ○ Send acknowledgments.



Two approaches to packet switching

1. Datagram Approach.
2. Virtual Circuit Approach.



Packet switching – Datagram approach

- ★ Datagram Packet Switching is also known as connectionless switching.
- ★ Each independent entity is called as datagram.
- ★ Datagrams contain destination information and the intermediary devices uses this information to forward datagrams to right destination.
- ★ In Datagram Packet Switching approach, the path is not fixed.
- ★ Intermediate nodes take the routing decisions to forward the packets.



Packet switching – Virtual Circuit approach

- ★ Virtual Circuit Switching is also known as connection-oriented switching.
- ★ In the case of Virtual circuit switching, a preplanned route is established before the messages are sent.
- ★ Call request and call accept packets are used to establish the connection between sender and receiver.
- ★ In this approach, the path is fixed for the duration of a logical connection.

